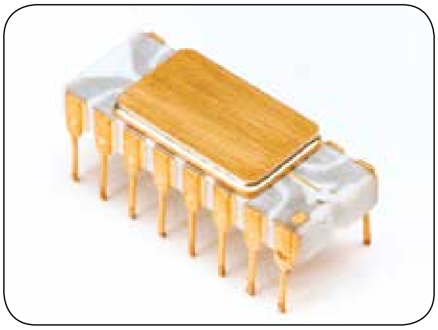
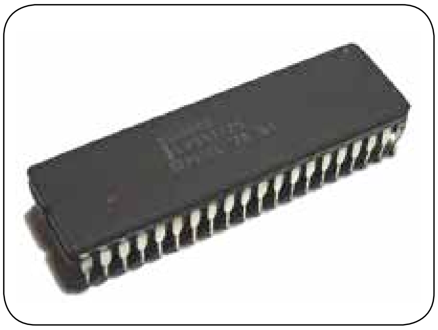
INTEL TIMELINE



Intel 4004 (1971)



Intel 8088 (1979)

Intel 486 (1989)

Intel Pentium 1 (1993)

Intel Pentium 4 HT (2004)

Intel Pentium Extreme 840 (2005)

Intel Core2Quad (2007)

Intel Core i7-4th Gen (2013)

niques are the future, and they drift away from silicon transistor-based computing. Still at a nascent stage, quantum computing replaces electronic binary-coded data with qubits. The quantum bits are quantum-mechanical representations of singular bits of information, preserved in two individually polarized states at the same time. The advantage of quantum computing over silicon-based computing lie in the quantum principles of functioning. While a single bit in present-day silicon computing can function only in the binary state (i.e., 0 or 1), a qubit can simultaneously exist in a superposition of two states (0 **and** 1).

Hence, an arrangement of x number of qubits will give rise to 2x bits of information, thereby processing data and gathering information at a much faster pace than any algorithm yet designed for the fastest multi-core processor. The quantum nature of this technology is far from the brute power increasing transistor rates, that have indeed made clock cycles faster, but now reached the horizontal slope of the developmental S-curve. Although quantum computing is yet to be finely developed, it is the future, and a need for faster computing alternative, that diverts technology away from Moore's half-century-old prediction.

The End

Present developments have come with indications that the future needs an alternative to silicon computing. Holographic displays are being made real, so it might not be long until Samantha from Her becomes reality. While such technology is indeed lucrative, it needs more horsepower than the current breed of commercially available multi-core processors can deliver.

Although the inertia of gradual development with transistors and cores will continue for a few years, the world of technology is taking a turn away from adding cores and pumping up clocks to boost performance. The prediction that Moore had made back in 1965 has inspired innovators to power up chipsets to deca-core processors in 2015. And even if it does not remain valid in future, you still can't deny its contribution towards shrinking large, box-shaped computers with chunky keys into to sleek, metal and glass touchscreen devices with many times more computing power than ever before.

Note: We have deliberately left supercomputers out of the equation, as the Moore's Law majorly pertains to technology that have powered devices in public sphere.